Department of Artificial Intelligence and Data Science (ADS)

Green University of Bangladesh

Semester: Spring 2026, B.Sc. in ADS

" Control Structures in Python "

Lab Report No: 03

Course Title: Introduction to Computational Thinking and Programming

Course Code: ADS 100

Section: 261 D2

Students Details

Name	ID
Naeem Shovon Shuvro	261035005

[For teachers use only: Don't write anything inside this box]

Lab Report Status	
Marks:	**Signature:**
Comments:	**Date:**

Lab Experiment Date: 14/03/ 26

Lab Submission Date: 27 /03/ 26

Course Teacher's Name: Wahia Tasnim

1. TITLE OF THE LAB REPORT EXPERIMENT

- **Write a python program to find the largest number between two numbers.**
- **Write a python program to find the smallest number between two numbers.**
- **Check if a number is even or odd.**
- **Write a python program to print the numbers from 1 to 10 using for loop and while loop.**
- **Print your name 10 times using for loop and while loop.**
- **Write a python program to find the sum of numbers from 1 to 10.**
- **Write a python program to find the even and odd numbers from 1 to 10.**

2. Objective/Aim:

The objective of this lab is to develop fundamental programming and problem-solving skills using Python by creating simple programs that utilize arithmetic operations, logical operators, conditional statements, loops, and user input/output.

Specifically, the aims are:

1. To write a Python program that determines the largest number between two given numbers using comparison operators and conditional statements.

2. To write a Python program that identifies the smallest number between two given numbers by applying comparison operator.

3. To write a Python program to check if a number is even or odd.

4. To print numbers from 1 to 10 using both for loop and while loop.

5. To print a name multiple times using loops.

6. To calculate the sum of numbers from 1 to 10 using loops.

7. To identify even and odd numbers within a given range using loops and conditions.

3. PROCEDURE:

The algorithm of the 6 problems is given below.

Problem 01: Finding the largest number between two numbers

Step 1: Start
Step 2: Take two integer numbers as input and store them in num1 and num2 variables respectively.
Step 3: Is num1 greater than num2? If yes, then go to step 4. If no, then go to step 5.
Step 4: Display the value of num1 as the largest.

Step 5: Is num1 less than num2? If yes, then go to step 6. If no, then go to step 7.
Step 6: Display the value of num2 as the largest.

Step 7: Display the value of num1 and num2 are equal.
Step 8: Stop

Problem 02: Finding the smallest number between two numbers

Step 1: Start
Step 2: Take two integer numbers as input and store them in num1 and num2 variables respectively.
Step 3: Is num1 smaller than num2? If yes, then go to step 4. If no, then go to step 5.
Step 4: Display the value of num1 as the smallest.

Step 5: Is num2 smaller than num1? If yes, then go to step 6. If no, then go to step 7.

Step 6: Display the value of num2 as the smallest.

Step 7: Display that both numbers are equal.

Step 8: Stop.

Problem 03: Checking if a number is even or odd

Step 1: Start
Step 2: Take an integer number as input and store it in variable num.
Step 3: Does num % 2 lead to 0? If yes, then go to step 4. If no, then go to step 5.
Step 4: Display num as an even number.
Step 5: Display num as an odd number.
Step 6: Stop.

Problem 04: Printing the numbers from 1 to 10 using loops.

Step 1: Start
Step 2: Declare a variable i and initialize it with value 1 in a loop.
Step 3: Is i smaller than 11? If yes, then go to step 4. If no, then go to step 6.
Step 4: Display the value of i with a space.
Step 5: Increase the value of i by 1 and go back to step 3.
Step 6: Stop.

Problem 05: Printing my name 10 times using loops.

Step 1: Start
Step 2: Declare a variable i and initialize it with value 1 in a loop.
Step 3: Is i smaller than 11? If yes, then go to step 4. If no, then go to step 6.
Step 4: Display my name "Shuvro" with a space.
Step 5: Increase the value of i by 1 and go back to step 3.
Step 6: Stop

Problem 06: Finding the sum of numbers from 1 to 10.

Step 1: Start
Step 2: Declare a variable sum1 with a value 0.
Step 3: Declare a variable i and initialize it with value 1 in a loop.
Step 4: Is i smaller than 11? If yes, then go to step 5. If no, then go to step 6.
Step 5: Set sum1 = sum1 + i and go back to step 4.
Step 6: Display the value of sum1.
Step 7: Stop.

Problem 07: Finding the even and odd numbers from 1 to 10.

Step 1: Start.

Step 2: Declare a varible i and initialize it with the value 1 in a loop.
Step 3: Is i smaller than 11? If yes, then go to step 4. If no, then go to step 6.
Step 4: If i%2 leads to 0, then display the value of i with a space as an even number.
Step 5: Increase the value of i by 1 and go back to step 3.
Step 6: Display a blank line.
Step 7: Declare a varible i and initialize it with the value 1 in another loop.
Step 8: Is i smaller than 11? If yes, then go to step 9. If no, then go to step 11.
Step 9: If i%2 doesn't lead to 0, then display the value of i with a space as an odd number.
Step 10: Increase the value of i by 1 and go back to step 8.
Step 11: Stop

4. TEST RESULT / OUTPUT:

Problem01:

Find the largest number between two numbers

Input:

```python
num1 = input("Enter first number: ")
num2 = input("Enter second number: ")

if num1 > num2:
    print(f"The largest number between {num1} and {num2} is {num1}")

elif num1 < num2:
    print(f"The largest number between {num1} and {num2} is {num2}")

else:
    print(f"Both {num1} and {num2} are equal")
```

Output:

```
uv run lab_03_p1_largest_number.py
Enter first number: 2
Enter second number: 4
The largest number between 2 and 4 is 4
```

Problem02:

Find the smallest number between two numbers:

Input:

```python
num1 = input("Enter first number: ")
num2 = input("Enter second number: ")

if num1 < num2:
    print(f"The smallest number between {num1} and {num2} is {num1}")
elif num2 < num1:
    print(f"The smallest number between {num1} and {num2} is {num2}")
else:
    print(f"Both {num1} and {num2} are equal")
```

Output:

```
uv run lab_03_p2_smallest_number.py
Enter first number: 30
Enter second number: 23
The smallest number between 30 and 23 is 23
```

Problem03:

Checking if a number is even or odd:

Input:

```python
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is a even number")
else:
    print(f"{num} is a odd number")
```

Output:

```
uv run lab_03_p3_check_even_odd.py
Enter a number: 5
5 is a odd number
```

Problem04:

Printing the numbers from 1 to 10 using loops:

Input:

```python
for i in range(1, 11):
    print(i, end=" ")
```

```python
i = 1
while i <= 10:
    print(i, end=" ")
    i += 1
```

Output:

```
uv run lab_03_p4_print_for_loop.py
1 2 3 4 5 6 7 8 9 10
```

Problem05:

Printing my name 10 time using loops:

Input:

```python
for i in range(1, 11):
    print("Shuvro", end=" ")
```

```python
i = 1
while i <= 10:
    print("Shuvro", end=" ")
    i += 1
```

Output:

```
uv run lab_03_p5_print_name_loop.py
Shuvro Shuvro Shuvro Shuvro Shuvro Shuvro Shuvro Shuvro Shuvro Shuvro
```

Problem06:

Finding the sum of numbers from 1 to 10:

Input:

```python
sum1 = 0
for i in range(1, 11):
    sum1 += i

print(f"Sum of 1 to 10 is: {sum1}")
```

Output:

```
uv run lab_03_p6_sum_of_1to10.py
Sum of 1 to 10 is: 55
```

Problem07:

Finding the even and odd numbers from 1 to 10:

Input:

```python
print("Odd numbers are:", end=" ")
for i in range(1, 11):
    if i % 2 != 0:
        print(i, end=" ")

print("\n")

print("Even numbers are:", end=" ")
for i in range(1, 11):
    if i % 2 == 0:
        print(i, end=" ")
```

Output:

```
uv run lab_03_p7_even_odd_1to10.py
Odd numbers are: 1 3 5 7 9

Even numbers are: 2 4 6 8 10
```

---

5. Key Python Concepts Used

During the implementation of these programs, several important Python programming concepts related to control structures were applied.

Variables:
Variables were used to store input numbers, intermediate results, and calculated values such as the largest number, smallest number, and the sum of odd and even numbers.

Conditional Statements (if–else):
Conditional statements were used to compare numbers and make decisions in the program. For example, the if statement was used to determine which number is larger or smaller between two numbers.

Comparison Operators:
Comparison operators such as >, <, and = = were used to compare values and determine logical conditions required for decision making.

Loops:
Loops were used to iterate through a set of numbers in order to calculate the sum of even and odd numbers. This helped process multiple values efficiently.

Arithmetic Operators:
Arithmetic operators such as + and % were used to perform calculations. The modulus operator % was particularly useful for determining whether a number is even or odd.

User Input and Output:
The functions input() and print() were used to take values from the user and display results in a clear format.

6. Conclusion

This laboratory experiment helped me understand how control structures in Python can be used to solve different logical and repetitive problems. By completing these exercises, I practiced using conditional statements, loops, comparison operators, and arithmetic operations.

In the first and second problems, I used conditional statements to compare two numbers and determine the largest and smallest values. These problems helped me understand how decision-making works in programming.

The third problem focused on checking whether a number is even or odd using the modulus operator. This helped me understand how conditions can be applied to classify numbers.

In the fourth and fifth problems, I used both for loops and while loops to print numbers and repeat tasks. These exercises showed how loops can be used to execute statements multiple times efficiently.

The sixth problem involved calculating the sum of numbers from 1 to 10 using a loop, which helped reinforce the concept of iteration and accumulation.

Finally, the seventh problem demonstrated how to separate even and odd numbers within a range using both loops and conditional statements.

Overall, this lab improved my understanding of Python control structures and strengthened my ability to write logical and structured programs. These concepts are essential for solving more complex programming problems in the future.